

Known at the origin o

Three orthogonal blocks

Error-weighted handover

Output

**Unit's own history**

hourly load, temperature  
and calendar before o



pseudo-origins o - 744k



completed forecasts give every  
candidate its own error record

**TimesFM 2.5**

zero-shot, load only  
1,344 h  $\rightarrow$  744 h, issued at  
o and at o - 744k ( $k \leq 6$ )

**Level  $\ell$** 

six historical candidates: recent, long-history  
and annual means, with and without weather

+ TimesFM window mean

weights  $\propto 1/\text{MSE}$  at pseudo-origins, shrunk to equal ( $K_0 = 8$ )

**Centred daily path  $b$** 

seven historical candidate paths: 8-, 4- and 2-week  
day types, the annual window, the weather response

same error weighting ( $K_0 = 2$ );  $\sum_d b_d = 0$

**Within-day shape  $w$** 

day-demeaned TimesFM trajectory  $w^T$

+ the unit's analog-day shape  $S$ : same day type,  
 $\pm 14$  days of year, up to 8 past days

$w = w^T + \omega_k (S - w^T)$ ,  $\omega_k \leq 1/2$  from the unit's errors  
at 3 pseudo-origins, shrunk to 0 (2.0)

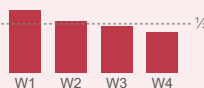
**Week-by-week  
handover**

$$m_d = (1 - \alpha_w) m_d^H + \alpha_w m_d^T$$

$m^H$ : history ( $\ell + b$ )

$m^T$ : TimesFM daily means

$\alpha_w$ : least squares on  $\leq 6$   
pseudo-origin pairs (fixed  
division vs TimesFM),  
shrunk to  $1/2$  ( $K_0 = 2$ )

**ANKYRA**

$$F_{d,h} = m_d + w_{d,h}$$

projected onto  $F \geq 0$

no training on the  
target series

**off / micro-load switch**

last 168 h  $\leq 10^{-6}$  kW, or  
all 1,344 h  $\leq 10^{-3}$  kW  
 $\rightarrow$  TimesFM unchanged

**Readouts**

energy 744 $\ell$

peak  $\max_d (\hat{L}_d + A_{r_d})$

interval pseudo-origin  
residual quantiles

Exact identity:  $\text{MSE} = \ell(e)^2 + \langle b_d(e)^2 \rangle_d + \langle w_{d,h}(e)^2 \rangle_{d,h}$  — the blocks are orthogonal, so each source is scored and anchored block by block (all three since 2.0).

from the unit's own history   from TimesFM 2.5   ANKYRA's error-weighted handovers